

Lab-3

Requirements Engineering – Elicitation

Student Placement System

Software Engineering — Lab Record

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KIET Deemed to be University

1. Stakeholder Identification

The stakeholders below were identified for the Student Placement System based on the actors involved in the end-to-end placement lifecycle, from company onboarding through to final selection and reporting. Each stakeholder's role, interest, and the elicitation technique used to gather their requirements are summarized in Table 1.

Stakeholder	Role / Interest	Elicitation Technique Used
Student	Primary end-user of the system; seeks accurate eligibility information, job listings, and transparent tracking of applications.	Interview, Questionnaire
Training & Placement Officer (TPO)	Owns and oversees the entire placement process; needs centralized control over drives, eligibility rules, and reporting.	Interview
Placement Coordinator	Assists the TPO with day-to-day coordination between students and recruiters; needs support for shortlisting and scheduling.	Interview, Observation
Recruiter / Company HR	External stakeholder posting job openings and reviewing candidates; needs filtered, verified student profiles.	Interview
Faculty Mentor	Guides assigned students academically; needs visibility into student eligibility and placement progress.	Questionnaire
College Management	Sponsors and oversees the placement cell; interested in overall placement statistics and institutional standing.	Interview
System Administrator	Maintains user accounts, access rights, and overall system configuration.	Observation
Development Team	Builds and maintains the system; requires clear, unambiguous, and testable requirements.	Observation

2. Reference: Use Case Diagram

The actors and use cases below define the scope of the Use Case Diagram for the Student Placement System. The diagram is intentionally kept compact, with four actors and their associated use cases, so that it remains practical to draw in a lab record.

Actor	Use Cases
Student	Register / Login, Manage Profile, Upload Resume, View Eligible Drives, Apply for Job, Track Application Status, View Interview Schedule, Receive Notifications
TPO	Login, Verify Student Eligibility, Approve Company Registration, Define Eligibility Criteria, Schedule Placement Drive, Shortlist Students, Schedule Interview/Test, Update Selection Status, Generate Placement Reports
Recruiter	Register / Login, Register Company, Post Job/Internship, View Shortlisted Profiles, Update Interview Results
Admin	Login, Manage User Accounts, Manage System Configuration, View Audit Logs

Include / Extend Relationships

- «include»: "Apply for Job" includes "Check Eligibility" — eligibility is always verified before an application is accepted.
- «include»: "Post Job/Internship" includes "Define Eligibility Criteria" — every posting must specify its eligibility rules.
- «extend»: "Register / Login" is extended by "Reset Password" — invoked only when a user forgets credentials.
- «extend»: "Track Application Status" is extended by "Receive Notifications" — a notification is an optional trigger, not a mandatory step.

3. Requirement Elicitation Techniques

Four complementary techniques were used to elicit requirements for the Student Placement System: interviews, a structured questionnaire, direct observation of the existing manual workflow, and user personas. Each technique surfaced a different dimension of the problem, and together they were used to cross-validate the requirements presented in Section 5.

3.1 Interviews

Semi-structured interviews were conducted separately with representatives of each major stakeholder group. Findings are summarized by stakeholder type below.

Student Interview Findings

- Placement announcements arrive through multiple, uncoordinated channels (notice board, WhatsApp groups, email), which has caused students to miss drives.
- There is no single place to check eligibility (CGPA, backlog count, branch) for a company before applying.
- Once an application is submitted, students have no visibility into its current stage.
- Resumes must be resubmitted separately for nearly every drive, since no reusable profile exists.

TPO Interview Findings

- Verifying student eligibility manually against spreadsheets is slow and error-prone, particularly for large batches.
- There is no consolidated dashboard showing drive-wise participation and outcome statistics.

- Shortlisting is performed manually by cross-checking company criteria against a master spreadsheet.
- Communicating interview schedules and results to students is fragmented across channels.

Placement Coordinator Interview Findings

- Coordinating between the TPO and recruiters for interview scheduling is largely manual, over calls and email.
- Historical placement records across academic years exist only as separate, disconnected files.
- It is difficult to distinguish, at a glance, students who withdrew from a drive from those still in process.

Recruiter Interview Findings

- Resumes are received in inconsistent formats over email, requiring manual filtering before shortlisting.
- There is no standard way to specify eligibility criteria before receiving applications, leading to unsuitable applicants.
- Sharing interview feedback and final results back to the college is currently an informal, ad hoc process.

3.2 Questionnaires

A short questionnaire of eight questions was circulated among final-year students to quantify the problems raised during interviews. A total of 120 responses were collected.

Questionnaire Items

- Q1. How do you currently receive placement-related notifications?
- Q2. Are you able to confirm your eligibility for a company before applying?
- Q3. Have you ever missed a placement drive due to delayed or missed information?
- Q4. How do you currently track the status of your submitted applications?
- Q5. Would a centralized placement portal significantly reduce your effort?
- Q6. How important is it to have your resume and profile stored at one place? (rate 1–5)
- Q7. Have you faced difficulty understanding which companies you are eligible for?
- Q8. Would you prefer real-time notifications for interview schedules and results?

Summarized Results (n = 120)

Question Topic	Response Summary
Notification channel used	Multiple uncoordinated channels: 62% WhatsApp: 24% Notice board: 9% Email: 5%
Able to confirm eligibility beforehand	Sometimes: 54% No: 31% Yes: 15%
Missed a drive due to delayed information	Yes: 46% No: 54%
Method of tracking application status	Follow up personally: 58% Informed by TPO: 33% No tracking: 9%
A centralized portal would reduce effort	Yes: 81% Not sure: 15% No: 4%
Importance of a single stored profile (avg. rating)	4.3 / 5
Difficulty identifying eligible companies	Yes: 63% No: 37%

3.3 Observation

The existing manual placement workflow was observed across one placement cycle. The typical sequence of activities is as follows:

Company Registration → Eligibility Criteria Definition → Job / Drive Posting → Student Notification → Eligibility Verification → Student Application → Shortlisting → Test / Interview Scheduling → Interview Conduct → Selection Status Update → Placement Record Update → Report Generation

Pain Points Observed

- Eligibility checking is done manually against a spreadsheet, introducing delay and occasional errors.
- Notifications are pushed through multiple channels, so some students consistently miss deadlines.
- Shortlisting requires the TPO to manually cross-reference resumes against criteria, which does not scale for large batches.
- No centralized dashboard exists for drive-wise statistics, so season-end reports are compiled manually.

From Observation to Requirements

- Manual eligibility checking → need for an automated eligibility-verification capability tied to each drive.
- Multi-channel, inconsistent notifications → need for a single, centralized notification mechanism within the system.
- Manual shortlisting → need for criteria-based shortlisting support.
- Manually compiled reports → need for a built-in reporting capability drawing on live placement data.

3.4 User Personas

Three personas were developed to keep the requirements grounded in realistic user needs throughout the design process.

Arya — Student Persona (Age 21)

Background: Final-year B.Tech (CS) student preparing for campus placements; relies on class WhatsApp groups and the notice board for updates.

Goals:

- Secure a suitable job offer before the final semester ends.
- Know in advance which companies she is eligible for.
- Track the status of each application without repeatedly asking the placement cell.

Frustrations with Current Process:

- Has missed a drive because the announcement was posted only in a WhatsApp group she had muted.
- Has to resubmit her resume separately for almost every company.
- Cannot tell whether her application has been shortlisted, rejected, or is still pending.

Vivek — TPO / Placement Officer Persona (Age 45)

Background: Training & Placement Officer with over a decade of experience, currently coordinating placements for more than 1,500 eligible students across branches.

Goals:

- Verify student eligibility quickly and accurately for each drive.
- Maintain a single dashboard of drive-wise statistics for management reviews.
- Reduce the manual effort spent compiling spreadsheets each season.

Frustrations with Current Process:

- Depends heavily on manually maintained Excel sheets that are easy to misplace or duplicate.
- Coordinating several recruiters at once, each with different timelines, is difficult without a shared system.
- Generating placement reports on short notice for management is time-consuming.

Sneha — Recruiter / Company HR Persona (Age 32)

Background: HR Manager at a mid-size IT company who visits 15–20 colleges each placement season to hire fresh graduates.

Goals:

- Receive a filtered list of eligible, verified candidates quickly.
- Post job requirements with clear eligibility criteria in one place.
- Communicate interview schedules and results efficiently to the college.

Frustrations with Current Process:

- Receives resumes in inconsistent formats by email, which slows down initial screening.
- Has no standard way to state eligibility criteria before applications arrive.
- Coordinating interview slots over phone calls and email is repetitive and error-prone.

4. User Stories

A total of 25 user stories were derived from the interviews, questionnaire results, observation, and personas above. They are organized by feature area, and each is written in the form: "As a <role>, I want to <action>, so that <benefit>."

4.1 Authentication

ID	As a...	I want to...	So that...	Priority
US-1	As a student,	I want to register and log in using my institutional credentials,	so that only verified students can access placement services.	High
US-2	As a TPO,	I want to log in with elevated access,	so that I can manage placement drives and student records securely.	High
US-3	As a recruiter,	I want to register and log in to a dedicated portal,	so that I can post jobs and review applicants.	High

4.2 Student Profile & Eligibility

ID	As a...	I want to...	So that...	Priority
US-4	As a student,	I want to create and update my academic and personal profile,	so that recruiters see accurate and current information.	High
US-5	As a student,	I want to upload and update my resume,	so that I do not need to resubmit it for every drive.	Medium
US-6	As a student,	I want to see which companies I am eligible for based on my academic record,	so that I only spend time applying where I qualify.	High
US-7	As a TPO,	I want to define eligibility criteria (CGPA, backlog,	so that only qualified students are allowed to apply.	High

		branch) for each drive,		
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4.3 Placement / Job Management

ID	As a...	I want to...	So that...	Priority
US-8	As a recruiter,	I want to register my company on the portal,	so that I can participate in placement drives.	High
US-9	As a recruiter,	I want to post job or internship openings with role details and criteria,	so that eligible students can view and apply to them.	High
US 10	As a TPO,	I want to review and approve company registrations,	so that only legitimate recruiters use the portal.	Medium
US 11	As a TPO,	I want to schedule placement drives with dates and venues,	so that students and coordinators are informed in advance.	High

4.4 Application & Shortlisting

ID	As a...	I want to...	So that...	Priority
US 12	As a student,	I want to apply to a job posting using my saved profile,	so that the application process is quick and consistent.	High

ID	As a...	I want to...	So that...	Priority
US 13	As a student,	I want to track the real-time status of my applications,	so that I know where I stand in each process.	High
US 14	As a TPO,	I want to shortlist students automatically based on defined eligibility criteria,	so that manual filtering effort is reduced.	High
US 15	As a recruiter,	I want to view a filtered list of eligible applicant profiles,	so that I can review candidates efficiently.	High

4.5 Placement Drive & Interview Management

ID	As a...	I want to...	So that...	Priority
US 16	As a TPO,	I want to schedule test and interview rounds for shortlisted students,	so that the selection process runs on time.	High
US 17	As a recruiter,	I want to update interview and selection results,	so that the college has accurate outcome records.	High
US 18	As a placement coordinator,	I want to view the list of students progressing through each round,	so that I can coordinate logistics effectively.	Medium

US 19	As a student,	I want to view my interview schedule and venue details,	so that I can prepare and attend on time.	High
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4.6 Notifications

ID	As a...	I want to...	So that...	Priority
US 20	As a student,	I want to receive notifications about new drives, shortlists, and results,	so that I do not miss important updates.	High
US 21	As a TPO,	I want to send announcements to specific eligible groups of students,	so that irrelevant notifications are avoided.	Medium

4.7 Reports & Administration

ID	As a...	I want to...	So that...	Priority
US 22	As a TPO,	I want to generate placement statistics and reports,	so that I can present placement performance to management.	High
US 23	As a member of college management,	I want to view overall placement statistics,	so that I can assess the institution's placement performance.	Medium
US 24	As a system administrator,	I want to manage user accounts and roles,	so that access to the system remains secure and appropriate.	Medium
US 25	As a system administrator,	I want to view system audit logs,	so that any irregular activity can be reviewed.	Low

5. Requirement Gathering Report

5.1 Purpose

The purpose of this report is to document the requirement elicitation process carried out for the Student Placement System and to present the resulting functional and non-functional requirements in a form suitable for subsequent design and development phases.

5.2 Elicitation Summary

Requirements were gathered using four techniques applied to eight identified stakeholder groups (Section 1). Interviews surfaced qualitative pain points specific to each stakeholder type; the questionnaire quantified how widespread these pain points were among students; observation of the existing manual workflow revealed where delays and errors were introduced; and personas kept the resulting requirements anchored to realistic user needs. Consistently, all four techniques pointed to the same underlying issues: fragmented communication, manual and error-prone eligibility checking, and a lack of centralized application tracking and reporting. These findings directly shaped the user stories in Section 4 and the requirements below.

5.3 Functional Requirements

ID	Requirement	Related Use Case(s)
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FR-1	The system shall allow students, TPO staff, and recruiters to register and authenticate using role-specific credentials.	Register / Login
FR-2	The system shall allow students to create and update a profile containing academic and personal details.	Manage Profile
FR-3	The system shall allow students to upload and update a resume document.	Upload Resume
FR-4	The system shall allow the TPO to define eligibility criteria (CGPA, backlog count, branch) for each placement drive.	Define Eligibility Criteria
FR-5	The system shall automatically determine and display eligible drives for each student based on the defined criteria.	View Eligible Drives, Check Eligibility
FR-6	The system shall allow recruiters to register their company profile on the portal.	Register Company
FR-7	The system shall allow the TPO to review and approve or reject company registrations.	Approve Company Registration
FR-8	The system shall allow recruiters to post job or internship openings with role details, eligibility criteria, and deadlines.	Post Job/Internship
FR-9	The system shall allow eligible students to apply to a job posting using their saved profile.	Apply for Job
FR-10	The system shall allow students to track the real-time status of each submitted application.	Track Application Status
FR-11	The system shall allow the TPO to shortlist applicants automatically based on defined eligibility criteria.	Shortlist Students
FR-12	The system shall allow recruiters to view filtered profiles of shortlisted candidates.	View Shortlisted Profiles
FR-13	The system shall allow the TPO to schedule test and interview rounds, including date, time, and venue.	Schedule Interview/Test
FR-14	The system shall allow recruiters to update interview outcomes and final selection status for each candidate.	Update Interview Results, Update Selection Status
FR-15	The system shall notify students of new drives, shortlisting outcomes, and interview schedules.	Receive Notifications

ID	Requirement	Related Use Case(s)
FR-16	The system shall allow the TPO to generate placement statistics and reports for a given period or drive.	Generate Placement Reports
FR-17	The system shall allow the system administrator to manage user accounts and access roles.	Manage User Accounts
FR-18	The system shall maintain an audit log of critical administrative actions.	View Audit Logs

5.4 Non-Functional Requirements

Category	Requirement
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Usability	A first-time student user shall be able to complete profile registration within 10 minutes without external assistance.
Security	The system shall encrypt stored passwords and enforce role-based access so that students cannot access TPO or administrator functions.
Performance	The system shall display a student's list of eligible drives within 3 seconds under normal load conditions.
Reliability	The system shall retain application and eligibility data without loss throughout an active placement season.
Availability	The system shall remain available at least 99% of the time during active placement drive periods, excluding scheduled maintenance.
Maintainability	The system shall be structured so that new eligibility rules or drive types can be added without major redevelopment.
Scalability	The system shall support simultaneous access by the entire eligible student batch during peak events such as shortlist announcements.
Auditability	The system shall log all administrative changes to eligibility criteria and student records with a timestamp and user identifier.

5.5 Assumptions and Constraints

- Students possess valid institutional login credentials issued by the college.
- Recruiters are responsible for the accuracy of the company and job information they provide.
- Academic records (CGPA, backlog status) are made available to the system by the institution's academic section.
- Continuous internet access is required to use the portal; no offline mode is assumed.
- Resume uploads are restricted to standard document formats such as PDF.
- The system facilitates and streamlines the placement process but does not guarantee student placement.
- Integration with external job portals or third-party recruitment platforms is outside the scope of this phase.
- The system is scoped to a single academic institution per deployment.

5.6 Conclusion

The combined use of interviews, a structured questionnaire, direct observation, and user personas allowed the requirement elicitation process for the Student Placement System to be both qualitative and quantitative. Interviews identified the specific difficulties faced by each stakeholder group, the questionnaire confirmed how common these difficulties were among the student population, observation traced the pain points to concrete steps in the existing manual workflow, and personas ensured that the derived requirements remained grounded in realistic user goals rather than an exhaustive but unfocused feature list. As a result, the 25 user stories, 18 functional requirements, and 8 categories of non-functional requirements presented in this report are mutually consistent and traceable back to evidence gathered directly from stakeholders.
